

PADDY YIELD PREDICTION

**BACHELOR OF TECHNOLOGY
IN**

COMPUTER SCIENCE AND ENGINEERING

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Kanuru, Vijayawada – 520007

(2026-27)

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CERTIFICATE

This is to certify that the project report title “**Paddy Yield Prediction**” is the bonafide work of **U.S.V.Rishi Raj (24501A05O4)**, **S.Mario Gnan (25505A0520)**, **Shaik Moulana (24501A05K8)**, **S.Pavani (24501A05M8)** in partial fulfilment of completing the Academic project in Data Mining Lab during the academic year 2026-27.

Signature of the course coordinator

Signature of the HOD

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1. Abstract

This project develops machine-learning models to predict the actual paddy (rice) yield, in kilograms, of a cultivated field from its agronomic and climatic conditions. The dataset contains field records from six agricultural blocks of Tamil Nadu, India. After removing 451 duplicate rows and six highly correlated columns, 2,338 records with 38 predictive features were used. A single reproducible pipeline was applied to every model: an 80:20 train–test split, IQR-based outlier clipping, StandardScaler and OneHotEncoder transformations, and SelectKBest feature selection. Nine regression algorithms — Linear, Ridge, Lasso, Decision Tree, Random Forest, Gradient Boosting, KNN, SVR and XGBoost — were tuned using GridSearchCV with 5-fold cross-validation and compared on the test set using R^2 , RMSE and MAE. XGBoost performed best, achieving an R^2 of 0.9904, an RMSE of about 905 Kg and an MAE of about 656 Kg. Exploratory analysis showed that Hectares is the strongest predictor of yield, while rainfall is a weak predictor. The final pipeline is deployed as an interactive Streamlit application for real-time single and batch predictions.

2. Introduction

2.1 Background

Agriculture remains the backbone of food security, and paddy (rice) is one of the most widely cultivated staple crops. Its productivity is shaped by a large number of interacting factors — rainfall, temperature, humidity, wind, soil type, seed rate, nursery method, and the timing and quantity of fertilizer and pesticide application. Because these variables interact in complex, non-linear ways, forecasting yield ahead of harvest is genuinely difficult with traditional rule-of-thumb methods.

Machine learning (ML) is well suited to this problem: given enough historical field records, ML models can learn the hidden relationships between growing conditions and the eventual yield, and then predict the yield of a new field from its conditions. This project applies that idea to real paddy-cultivation data from Tamil Nadu, India.

2.2 Motivation

- **Farm-level planning:** a reliable pre-harvest yield estimate lets farmers plan irrigation, fertilizer use and labour, and negotiate storage or sale in advance.
- **Policy & supply-chain decisions:** agricultural planners and policymakers can anticipate regional production, plan procurement and manage food-supply risk.
- **Insight into drivers:** feature-importance analysis reveals which agronomic and climatic factors most influence yield, guiding better agronomic practices.
- **Educational value:** the project provides end-to-end, hands-on practice in data cleaning, EDA, preprocessing pipelines, model tuning and deployment.

2.3 Problem Statement and Goal

Problem statement. Paddy productivity is highly influenced by many environmental and agronomic factors, and predicting yield levels in advance remains a major challenge because of the complex interaction between these variables.

Goal. To develop a machine-learning model that predicts the actual paddy yield (in kilograms) from historical agronomic and climatic data, and to deploy the best model as an easy-to-use application that supports informed decisions on irrigation planning, fertilizer usage and resource allocation.

3. Project Design

The end-to-end workflow of the project is summarised in Figure 3.1. It proceeds from raw data collection through cleaning, exploratory analysis, a preprocessing pipeline, model training and tuning, evaluation and selection, and finally deployment of the best model.



Figure 3.1 — Proposed project design (workflow).

4. Dataset Overview

The dataset contains real paddy-cultivation field records gathered from six agricultural blocks in Tamil Nadu, India (Cuddalore, Kurinjipadi, Panruti, Kallakurichi, Sankarapuram and Chinnasalem). Each row corresponds to one cultivated field and records the land and crop profile, the fertilizer/pesticide inputs applied, the weather across the growing season, and the final measured yield.

4.1 Size and Target Variable

- **Records:** 2,789 raw rows → 2,338 after removing 451 exact duplicate rows.
- **Columns:** 45 raw fields → 38 model features after dropping the target and six highly-correlated columns (see Section 5).
- **Target:** Paddy_yield_in_Kg — the total paddy yield of the field in kilograms; this is the value the models predict.
- **Source:** agricultural field data, Tamil Nadu, India.

4.2 Field and Agronomic Features

Table 4.1 lists the land, crop and input features recorded for each field.

Table 4.1 — Field and agronomic features.

Column	Description
Hectares	Total land area cultivated (hectares)
Agriblock	Agricultural block / region (6 categories)
Variety	Paddy variety — CO_43, ponmani, delux ponni
Soil_Types	Soil type — alluvial or clay
Seedrate_in_Kg	Quantity of seed used (Kg)
Nursery	Nursery method — dry or wet
Urea_40Days	Urea fertilizer applied at 40 days (Kg)
Potassh_50Days	Potash fertilizer applied at 50 days (Kg)
Pest_60Day_in_ml	Pesticide applied at 60 days (ml)
Trash_in_bundles	Crop residue / trash (bundles)

4.3 Weather Features

Weather is recorded across four 30-day growth stages — D1–D30 (establishment), D31–D60 (tillering), D61–D90 (flowering) and D91–D120 (grain filling). Each variable in Table 4.2 is measured once per stage, giving four columns per variable.

Table 4.2 — Weather features (measured per growth stage).

Column (per stage)	Description
*Rain__in_mm / *AI_in_mm	Rainfall and irrigation (mm) in the interval
Min_temp / Max_temp	Minimum and maximum temperature (°C)
Inst_Wind_Speed_..._in_Knots	Instantaneous wind speed (knots)
Wind_Direction	Prevailing wind direction (e.g. SW, NE, NNW)
Relative_Humidity	Relative humidity (%)

4.4 Categorical Feature Categories

Table 4.3 — Categorical features and their categories.

Feature	No. of Categories	Values
Agriblock	6	Cuddalore, Kurinjipadi, Panruti, Kallakurichi, Sankarapuram, Chinnasalem
Variety	3	CO_43, ponmani, delux ponni
Soil_Types	2	alluvial, clay
Nursery	2	dry, wet
Wind_Direction_D*	5–6	Compass directions per growth stage (e.g. SW, W, NE, NNW)

4.5 Data Characteristics

- The data mixes numeric features (area, inputs, temperatures, rainfall, humidity, wind speed) with categorical features (block, variety, soil, nursery, wind direction).
- The target yield is right-skewed, with most fields between 15,000 and 35,000 Kg and a median near 25,000 Kg.
- Several fertilizer-related columns are highly correlated with one another (multicollinearity), which is addressed during preprocessing.

5. Data Preprocessing

Raw agricultural data contains inconsistent column names, duplicate rows, redundant columns and occasional outliers. Preprocessing turns this raw table into clean, uniformly-scaled numeric inputs suitable for regression. Crucially, every step that learns from the data (outlier limits, scaling, encoding, feature selection) is fit on the training set only and then applied to the test set, so no information leaks from test to train.

5.1 Cleaning Column Names

Column headers were standardised: leading/trailing spaces were stripped and special characters (spaces, parentheses, slashes and hyphens) were replaced with underscores. This yields consistent, code-friendly names such as 30DRain__in_mm and Inst_Wind_Speed_D1_D30_in_Knots.

5.2 Missing Values and Duplicate Rows

The data was checked for null values, and 451 exact duplicate rows were found and removed, reducing the dataset from 2,789 to 2,338 unique records. De-duplicating before the train–test split is important: it prevents the same field appearing in both the training and test sets, which would otherwise inflate the measured accuracy.

5.3 Removing Multicollinear Columns

A correlation analysis (Section 6.3) revealed a block of fertilizer- and nursery-related columns that are almost perfectly correlated (correlation > 0.9) with one another and with retained features. Keeping all of them adds redundancy and destabilises linear models, so the following six columns were dropped:

- LP_Mainfield_in_Tonnes, LP_nurseryarea_in_Tonnes, Nursery_area__Cents
- DAP_20days, Weed28D_thiobencarb, Micronutrients_70Days

The most informative agronomic features — Hectares, Seedrate_in_Kg, Urea_40Days, Potassh_50Days, Pest_60Day_in_ml and Trash_in_bundles — together with all weather features were retained, leaving 38 predictive features.

5.4 Handling Outliers (IQR Clipping)

Numeric features were treated for outliers using the Inter-Quartile Range (IQR) method. For each numeric column the first and third quartiles (Q1, Q3) are computed on the training data and values are clipped to the range below; the same limits are then applied to the test data:

$$\text{lower} = Q1 - 1.5 \times IQR \quad \text{upper} = Q3 + 1.5 \times IQR, \quad \text{where } IQR = Q3 - Q1$$

Clipping (rather than deleting) keeps every field while limiting the influence of extreme values on scaling and on distance/kernel-based models.

5.5 Encoding and Scaling (ColumnTransformer)

Because regression algorithms require numeric input on comparable scales, a single ColumnTransformer applies two transformations in parallel:

- **StandardScaler** on numeric columns — rescales each feature to zero mean and unit variance, so that features measured in different units (Kg, mm, °C, knots) contribute comparably. Each value is transformed as:

$$z = (x - \mu) / \sigma$$

where x is the original value, μ the feature mean and σ its standard deviation.

- **OneHotEncoder** on categorical columns (Agriblock, Variety, Soil_Types, Nursery and the four Wind_Direction columns) — converts each category into binary indicator columns, avoiding any false ordinal relationship between categories.

5.6 Feature Selection (SelectKBest)

After encoding, SelectKBest with the `f_regression` score keeps the `k` features most strongly associated with the target. The value of `k` is itself tuned by the grid search (values such as 10, 15 and 20 were explored per model), so each algorithm uses the number of features that maximises its cross-validated performance.

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution of the target, the skewness and relationships of the features, and the correlations that motivated dropping redundant columns.

6.1 Target Variable Distribution

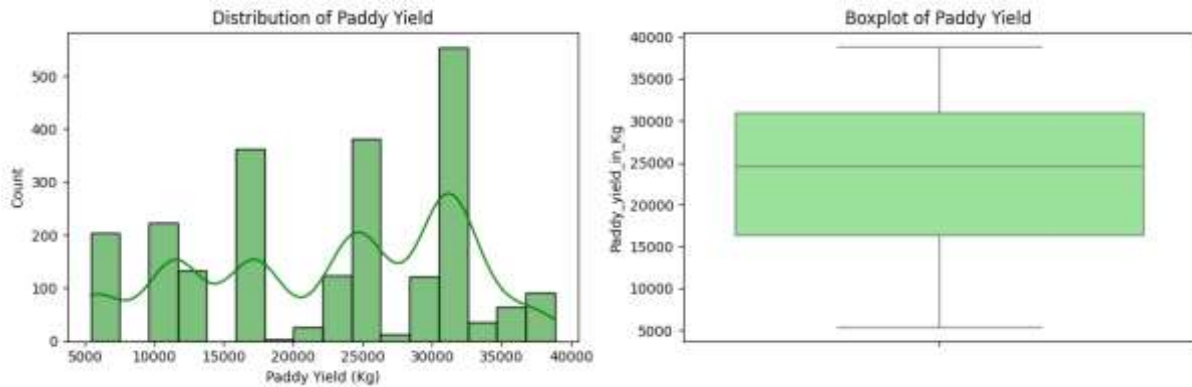


Figure 6.1 — Distribution and boxplot of paddy yield (Kg).

- The distribution of paddy yield is **right-skewed** — most fields produce between 15,000 and 35,000 Kg, with a peak around 30,000–35,000 Kg.
- The boxplot shows a **median yield of roughly 25,000 Kg**.
- A long lower whisker indicates some low-yield fields ($\approx 5,000$ Kg), likely due to poor soil, small area or adverse weather. No extreme outliers are visible in the target — it is relatively clean.

6.2 Skewness of Numerical Features

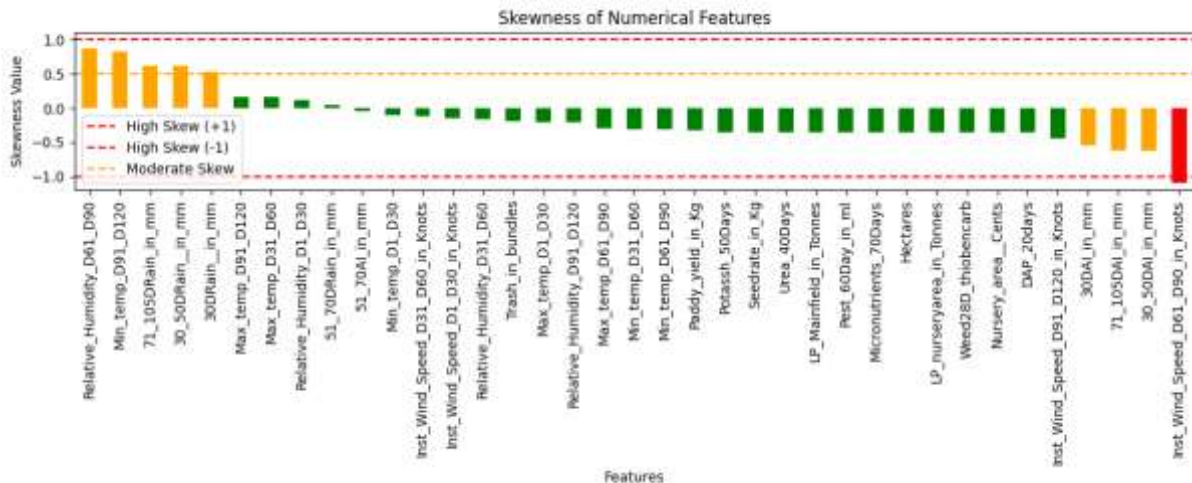


Figure 6.2 — Skewness of the numerical features.

- Most features show **moderate skewness** (0.5–1.0), which is acceptable for StandardScaler; *Relative_Humidity_D61_D90* and *Min_temp_D91_D120* are the most skewed.
- No feature crosses the ± 1 threshold, so no log transformation is required — StandardScaler is the appropriate choice.

6.3 Correlation Heatmap

The correlation matrix (Figure 6.3) shows how strongly features relate to each other and to the target, and is used to detect multicollinearity.

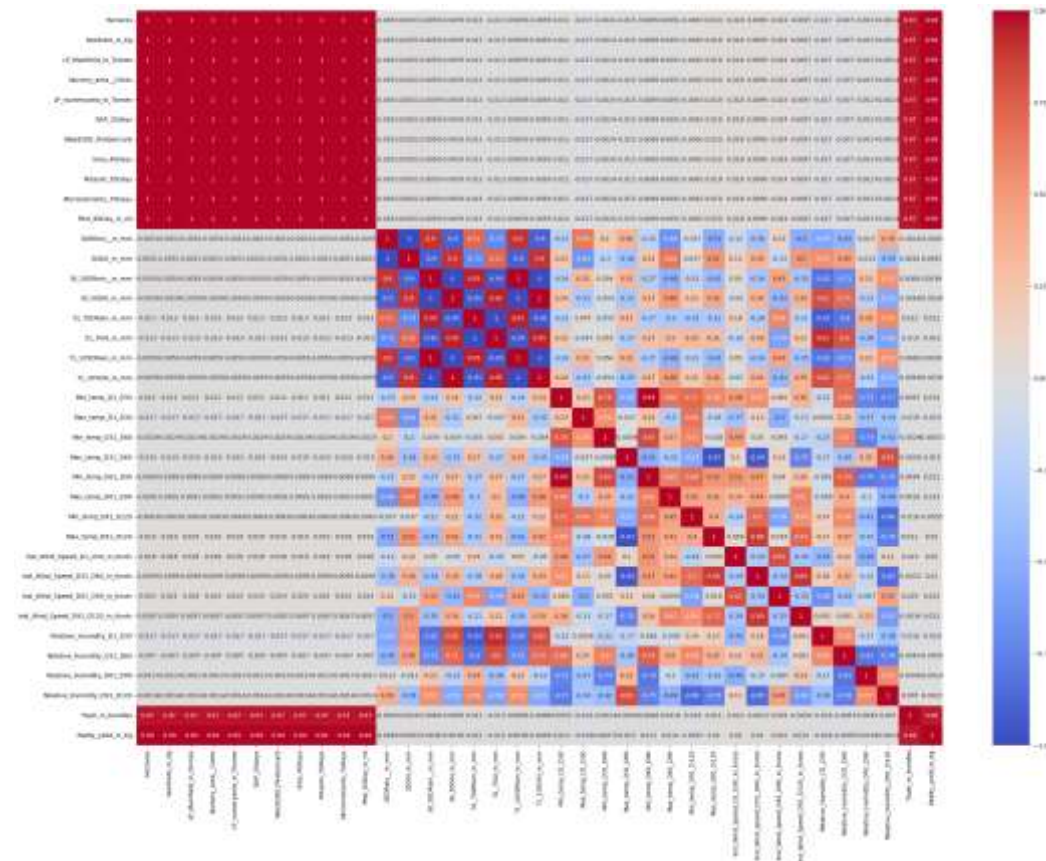


Figure 6.3 — Correlation heatmap of the numerical features.

- A block of fertilizer-related columns (LP_Mainfield, LP_nurseryarea, Nursery_area, DAP_20days ...) is **very highly correlated (> 0.9)** — these were dropped to remove redundancy.
- **Hectares shows a strong positive correlation with yield**, confirming it is the most important predictor.
- Individual weather features show only weak-to-moderate correlation with yield — no single weather variable dominates; their combined effect matters.

6.4 Feature–Yield Relationships

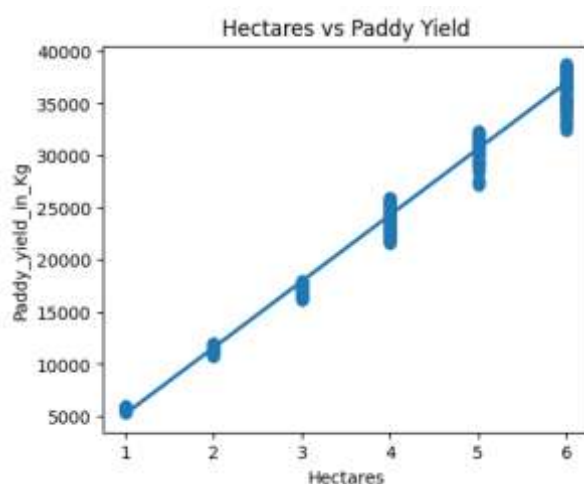


Figure 6.4 — Hectares vs Yield (regression).

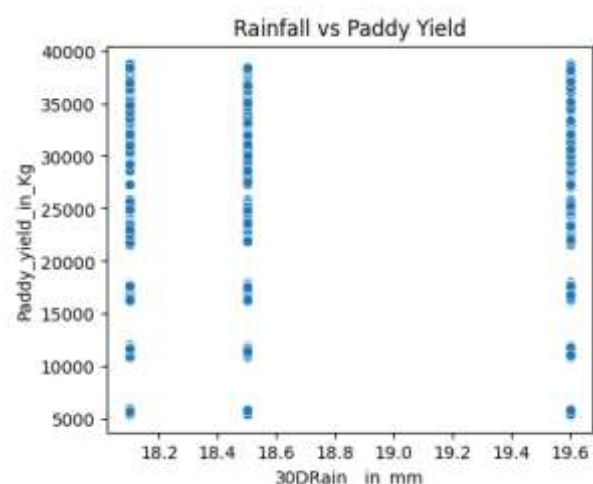


Figure 6.5 — Rainfall vs Yield (scatter).

- **Hectares vs Yield:** a strong positive linear relationship — as area rises from 1 to 6 hectares, yield rises proportionally. Hectares is the single strongest predictor.
- **Rainfall vs Yield:** rainfall clusters in a narrow range (≈ 18 – 20 mm) while yield varies widely — a vertical scatter that confirms rainfall alone is a weak predictor.

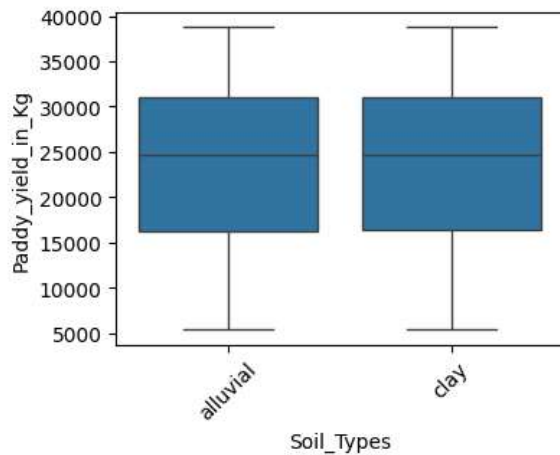


Figure 6.6 — Soil type vs Yield (boxplot).

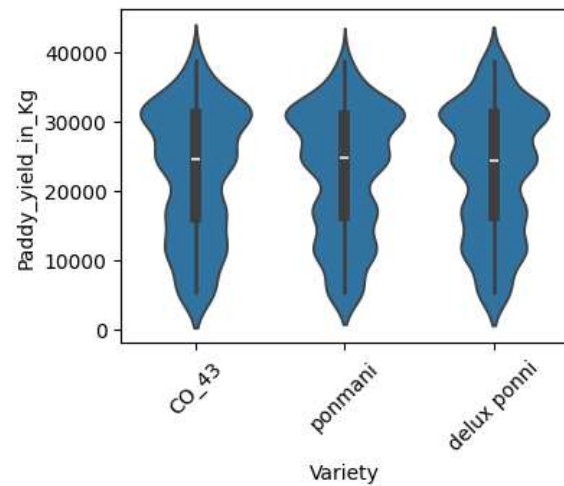


Figure 6.7 — Variety vs Yield (violin).

- **Soil type:** alluvial and clay soils show similar, overlapping yield distributions; alluvial has a slightly higher median (it is more fertile and water-retentive).
- **Variety:** CO_43, ponmani and delux ponni have broadly similar yield distributions — variety alone does not strongly differentiate yield; inputs and weather matter more.

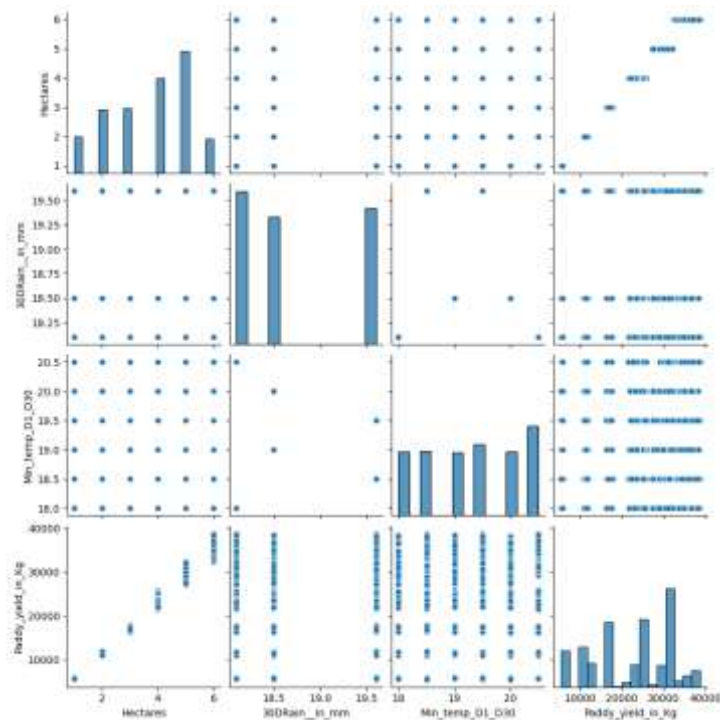


Figure 6.8 — Pair plot: Hectares, Rainfall, Min-temp vs Yield.

- The pair plot reinforces the findings: among these features, **Hectares is the only one with a clear direct relationship to yield**, while rainfall and temperature show limited range and no clear trend.

7. Model Training and Hyperparameter Tuning

After preprocessing, the data was split 80:20 into training and test sets (1,870 training and 468 test records). Nine regression algorithms were then trained on the identical pipeline so that their results are directly comparable.

7.1 Training Strategy

- **Train–test split:** 80:20 (random_state = 42) → 1,870 train / 468 test rows.
- **Pipeline (fixed for every model):** ColumnTransformer (StandardScaler + OneHotEncoder) → SelectKBest(f_regression) → estimator.
- **Cross-validation:** 5-fold cross-validation on the training set.
- **Grid search:** GridSearchCV searches each model's hyperparameters (and the number of selected features k), scoring on R^2 , and refits the best configuration.

7.2 Regression Algorithms

Linear, Ridge and Lasso Regression. Linear regression fits a weighted sum of the input features:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where $\hat{y}$ is the predicted yield, x_i the features, β_i the learned coefficients and β_0 the intercept. Ridge adds an L2 penalty (controlled by α) and Lasso an L1 penalty on the coefficients, which shrink and stabilise them and, for Lasso, drive some to zero — both help when features are correlated.

K-Nearest Neighbours (KNN) Regressor. A non-parametric model that predicts the average target of the K most similar fields in feature space:

$$\hat{y} = (1 / K) \cdot \sum y_i \quad (\text{averaged over the } K \text{ nearest neighbours of } x)$$

Small K is sensitive to noise; large K over-smooths. The best value found was $K = 7$.

Support Vector Regressor (SVR). Fits a function that stays within an ε -insensitive margin of the data, penalising only errors larger than ε :

$$f(x) = w^T x + b$$

Key hyperparameters are C (regularisation), the kernel, and ε (margin). A linear kernel with $C = 10$ performed best here.

Decision Tree, Random Forest and Gradient Boosting. A decision tree splits the feature space into regions and predicts a constant per region. Random Forest averages many trees trained on random subsets, reducing variance:

$$\hat{y} = (1 / T) \cdot \sum h_t(x) \quad (T \text{ trees, } h_t \text{ the } t\text{-th tree})$$

Gradient Boosting instead builds trees sequentially, each new tree correcting the errors of the current ensemble.

XGBoost (eXtreme Gradient Boosting). A high-performance gradient-boosting framework. It adds trees sequentially by gradient descent on the loss and includes L1/L2 regularisation to control complexity. The final prediction sums all trees:

$$\hat{y} = \sum f_k(x) \quad (\text{sum over } K \text{ boosted trees } f_k)$$

Key hyperparameters are the number of trees ($n_estimators$), tree depth (max_depth) and $learning_rate$.

7.3 Best Hyperparameters (per model)

GridSearchCV with 5-fold cross-validation produced the best configurations given in Table 7.1.

Table 7.1 — Best hyperparameters for each model.

Model	k	Best hyperparameters
Linear Regression	10	— (no additional hyperparameters)
Ridge	10	alpha = 0.01
Lasso	10	alpha = 1
Decision Tree	10	max_depth = 5, min_samples_split = 2
Random Forest	10	n_estimators = 100, max_depth = 5
Gradient Boosting	10	n_estimators = 100, learning_rate = 0.1
KNN	15	n_neighbors = 7
SVR	20	kernel = linear, C = 10
XGBoost	20	n_estimators = 100, max_depth = 3, learning_rate = 0.1

8. Model Evaluation and Selection

Each tuned model was evaluated on the untouched 468-row test set. Three complementary regression metrics were used.

8.1 Evaluation Metrics

R² Score (coefficient of determination) — the proportion of variance in yield explained by the model. 1.0 is a perfect fit; 0.0 is no better than predicting the mean.

$$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 \div \Sigma(y_i - \bar{y})^2]$$

Root Mean Squared Error (RMSE) — the square root of the mean squared error, in the same units as the target (Kg). It penalises large errors heavily; lower is better.

$$RMSE = \sqrt{(1 / N) \cdot \Sigma(y_i - \hat{y}_i)^2)}$$

Mean Absolute Error (MAE) — the average absolute error in Kg; robust and easy to interpret.

$$MAE = (1 / N) \cdot \Sigma | y_i - \hat{y}_i |$$

where y_i is the actual yield, $\hat{y}_i$ the predicted yield, $\bar{y}$ the mean actual yield and N the number of test records.

8.2 Results — All Models

Table 8.1 reports the test-set performance of all nine tuned models, sorted by R². The best value in each column belongs to XGBoost (highlighted), and Figure 8.1 compares the models graphically.

Table 8.1 — Test-set performance of all nine tuned models.

Model	Test R ²	RMSE (Kg)	MAE (Kg)
XGBoost	0.9904	905	656
Random Forest	0.9903	914	659
Gradient Boosting	0.9902	915	659
Decision Tree	0.9902	915	660
KNN	0.9890	970	701
Lasso	0.9878	1,022	763
Linear Regression	0.9878	1,022	763
Ridge	0.9878	1,022	763
SVR	0.9864	1,080	775

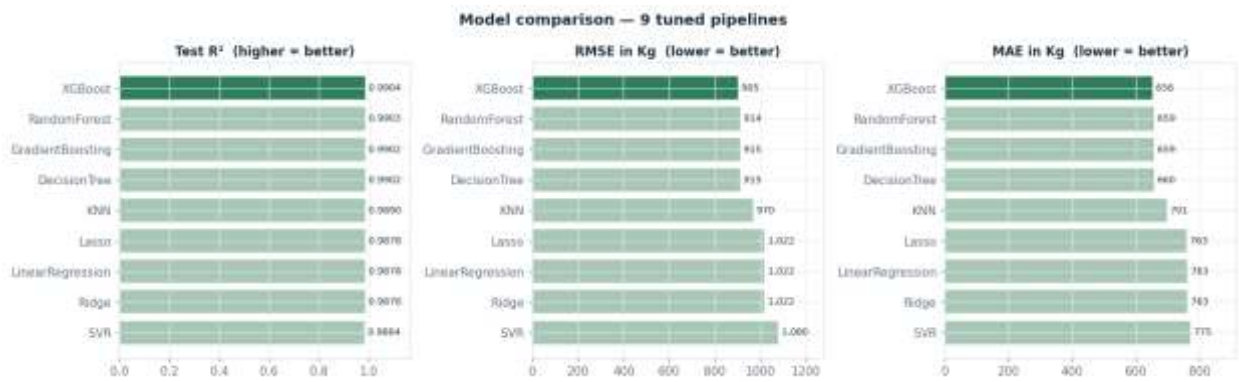


Figure 8.1 — Model comparison across Test R², RMSE and MAE (XGBoost wins on every metric).

8.3 Best-Model Diagnostics

For the winning XGBoost pipeline, the actual-vs-predicted plot (Figure 8.2) lies tightly along the diagonal, and the feature-importance plot (Figure 8.3) confirms the drivers of yield.

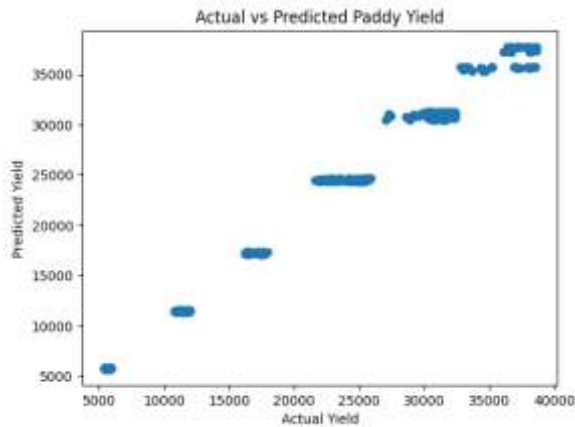


Figure 8.2 — Actual vs predicted yield (XGBoost).

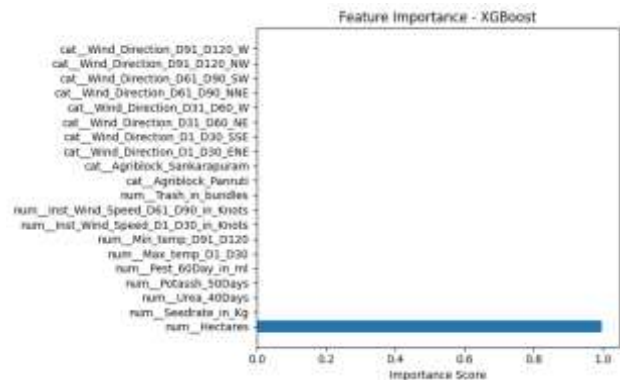


Figure 8.3 — Feature importance (XGBoost).

- Predictions closely track actual yield across the whole range, consistent with the $\approx 0.99 R^2$.
- **Hectares** dominates the importance ranking, confirming cultivated area as the primary driver of total yield, with agronomic inputs and weather contributing secondary effects.

8.4 Model Selection

The **XGBoost Regressor** is selected as the final model. It records the **highest R^2 (0.9904)** and the **lowest RMSE (≈ 905 Kg)** and **MAE (≈ 656 Kg)** of all nine candidates. Its strength comes from its ensemble, gradient-boosted nature and built-in regularisation, which capture the non-linear interactions between area, inputs and weather while resisting overfitting. The differences between the top tree-based models are small, but XGBoost is consistently best and is saved as *paddy_yield_model.pkl* for deployment.

9. Deployment — Interactive Application

The trained pipeline is deployed as an interactive Streamlit web application (“Rice Yield Predictor”) so that non-technical users can obtain predictions without writing code. The saved pipeline (paddy_yield_model.pkl) is loaded once and reused for every prediction. The app is organised into four pages.

9.1 Predict Yield

Users enter a farm profile (area, agriblock, variety, soil, nursery method, and seed/fertilizer/pesticide inputs) and the seasonal weather for each of the four 30-day growth stages via tabs. A live result panel updates instantly with:

- the predicted paddy yield in Kg and the yield per hectare;
- a 95% confidence band (approximately $\pm 2 \times \text{RMSE}$ around the prediction);
- a HIGH / MEDIUM / LOW yield category with a short agronomic recommendation.

9.2 Batch Predictions

A batch mode accepts a CSV with one row per field. The app validates that all 38 required feature columns are present, predicts the yield for every row together with lower/upper confidence bounds, shows summary metrics (total fields, mean and maximum yield), and offers the results as a downloadable CSV.

9.3 Model Info

This page exposes the modelling work behind the app. Users can switch the active model among all nine saved pipelines at runtime and inspect its R^2 , RMSE and MAE, a candidate-benchmark table across all models, a feature-importance chart, a residual-analysis plot on the recomputed holdout set, and the saved training diagnostics generated by the notebook.

9.4 Guide and About

A Guide page explains how to use the predictor and how to prepare batch CSV files, and an About page documents the problem, goal, dataset and key findings — making the application self-describing for new users.

10. Conclusion

10.1 Summary

This project built a complete, reproducible machine-learning workflow for predicting paddy yield from agronomic and climatic field data. After cleaning the data (standardising columns, removing 451 duplicates and dropping six multicollinear columns), performing EDA, clipping outliers and building a scaling/encoding/feature-selection pipeline, nine regression models were trained and tuned with 5-fold GridSearchCV. The XGBoost Regressor emerged as the best model, explaining about 99% of the variance in yield ($R^2 = 0.9904$, $RMSE \approx 905$ Kg, $MAE \approx 656$ Kg).

10.2 Key Findings and Business Impact

- **Hectares** (cultivated area) is the strongest predictor of total yield; agronomic inputs and stage-wise weather provide meaningful secondary signal.
- Rainfall varies little across the records and is a weak individual predictor; soil type and variety have only modest effects.
- The model lets farmers estimate expected yield before harvest, supporting better planning of irrigation, fertilizer use and resource allocation, and can feed into agricultural-advisory apps or government crop-planning systems.

10.3 Future Scope

- Incorporate additional data sources such as satellite/remote-sensing indices (e.g. NDVI), soil-nutrient tests and finer-grained daily weather.
- Explore deep-learning and sequence models (e.g. LSTMs) to capture temporal weather dynamics across the growing season more richly.
- Add prediction intervals from quantile or probabilistic models, and extend the app with multi-season and multi-region support.

11. References

Dataset

Paddy-cultivation field records (agronomic and climatic features), Tamil Nadu, India — used to train and evaluate the models in this project.

Tools and Libraries

- scikit-learn 1.9.1 — pipelines, preprocessing, model selection and metrics. <https://scikit-learn.org/stable/>
- XGBoost 3.4.1 — gradient-boosted tree regressor. <https://xgboost.readthedocs.io/>
- Streamlit — interactive web-application framework for deployment. <https://docs.streamlit.io/>
- pandas, NumPy, Matplotlib, seaborn, Altair — data handling and visualisation.

GITHUB LINKS

24501A05O4	-	https://github.com/rishirajunguturi-bit/Rice-Yeild-Predictor
25505A0520	-	https://github.com/mariognan13-lab/Rice-yield-predictor
24501A05K8	-	https://github.com/shaikmubasheer006/Rice-yield-predictor
24501A05M8	-	https://github.com/PavaniSykam/Rice-yield-predictor